

Blocks in Jharkhand is another step in building a multi-mineral profile for the company. In the future, NMDC intends to diversify its operations into new age minerals like Lithium and Nickel. According to Deb, the demand for these strategic minerals will rapidly grow in the upcoming years and therefore NMDC is mining for opportunities of overseas investments and project acquisitions for this diversification.

Digital Initiatives

Mr. Deb's vision has instilled confidence in NMDC's technology up-gradation initiatives across its projects for quantum increase in output and improved overall efficiency. As the company expands, it is making massive investments to enhance its technical and digital strength. NMDC is using some of the world's most powerful geospatial data analytics and mapping technologies for mineral exploration. Recently, NMDC collaborated with IIT Kharagpur for a first-of-its-kind initiative of drone-based mineral exploration.

NMDC is not only adopting cutting-edge technologies such as ERP through SAP HANA, Vision Enhancement System and Automation of Subsystems, the company is now in the process of implementing Fleet Management and contemplating the implementation of Automatic Sampling System at its mining complexes. Deb believes that interventions of this kind will bring digital disruption to the mining sector and drive NMDC to a 100 million tonnes mining capacity company by 2030. The future of mining will be driven by technology that can achieve the vision of Industry 4.0. NMDC is implementing these technologies and increasing investment in digitalisation based on their feasibility in the company's ecosystem.

R&D Capabilities

NMDC boasts of its own R&D Centre at Hyderabad which is recognized as a Centre of Excellence by UNIDO and also has ISO & EMS accreditations. The expertise of NMDC's R&D wing is not only used for benefitting and upscaling its own operations but also helps various sectoral clients like SAIL, CIL, and NTPC among others. The R&D Centre has an inheritance of over five decades serving the mineral industries and has evolved with time to extend its expertise. The Centre has subsequently grown into a strong, full-fledged R&D wing extending technology support to NMDC's in-house projects and other organizations in India and abroad.

The PSE's R&D Centre provides integrated solutions focusing on the life cycle of iron ore and other minerals under one roof. It is equipped with state-of-the-art testing facilities and a multi-disciplinary team of Geologists, Chemists and Mineral, Metallurgical, Electrical, Mechanical and Chemical Engineers to address every challenge. A strong Research and Development wing plays a key role in sustainable growth of an organization and today's NMDC is an iron-strong witness to it.

Sustainable Mining Practices

All the mining complexes of NMDC have been endowed with 5 Star rating by the Indian Bureau of Mines, Ministry of Mines, thanks to its scientific and sustainable mining practices. After planting over 3.5 million trees, setting up Solar Power Projects and Windmills for renewable power generation, NMDC is working towards 'Zero-Waste Mining' by exploring various grades of iron ore. Under the sustainability development programme, several Sewage Treatment Plants and Tailing Dams have been constructed at all its major projects. Most of their mining complexes have been awarded Greentech Environment Awards during the Mines Environment and Mineral Conservation Week year on year. Annual Carbon Footprint studies are being conducted for disclosure of Greenhouse Gas Emissions in the Carbon Disclosure Project (CDP). In the last decade, the company has reduced its total per tonne GHG emission intensity by 24%.

NMDC is conducting the Field Studies for reviewing its